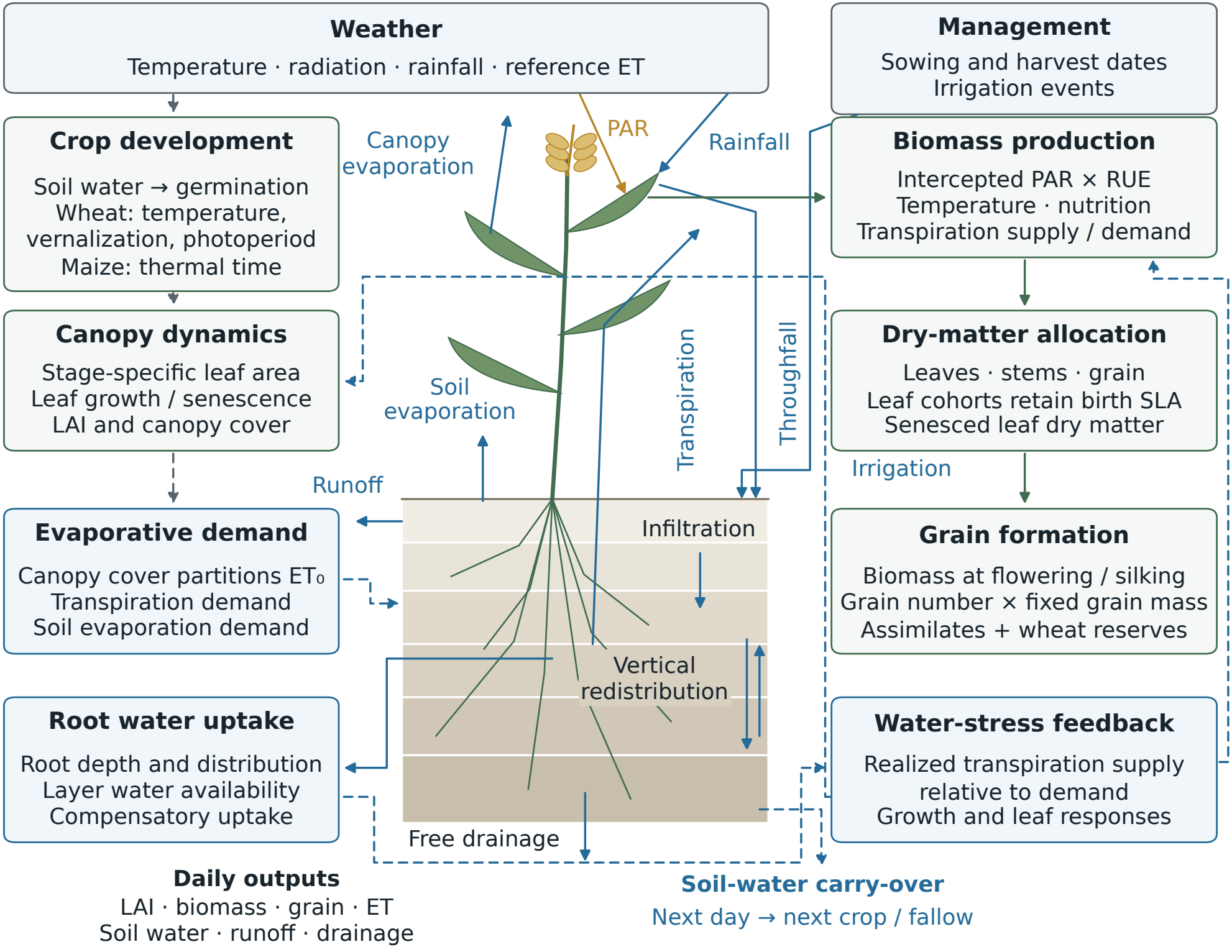
Weather
Temperature · radiation · rainfall · reference ET
Management
Sowing and harvest dates
Irrigation events
Crop development
Soil water → germination
Wheat: temperature, vernalization, photoperiod
Maize: thermal time
Canopy evaporation
PAR
Rainfall
Biomass production
Intercepted PAR × RUE
Temperature · nutrition
Transpiration supply / demand
Canopy dynamics
Stage-specific leaf area
Leaf growth / senescence
LAI and canopy cover
Soil evaporation
Transpiration
Throughfall
Dry-matter allocation
Leaves · stems · grain
Leaf cohorts retain birth SLA
Senesced leaf dry matter
Runoff
Irrigation
Evaporative demand
Canopy cover partitions ET₀
Transpiration demand
Soil evaporation demand
Infiltration
Grain formation
Biomass at flowering / silking
Grain number × fixed grain mass
Assimilates + wheat reserves
Vertical redistribution
Root water uptake
Root depth and distribution
Layer water availability
Compensatory uptake
Water-stress feedback
Realized transpiration supply relative to demand
Growth and leaf responses
Free drainage
Daily outputs
LAI · biomass · grain · ET
Soil water · runoff · drainage
Soil-water carry-over
Next day → next crop / fallow